

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<thought>

To solve this problem , we need to understand the symptoms and the diagnosis . The patient has a sore throat , non-productive cough , fever , and patchy reticular opacities on the X-ray , which suggests a community -acquired pneumonia . The most common pathogens causing community -acquired pneumonia are bacteria like Streptococcus pneumoniae , Staphylococcus aureus , and Mycoplasma pneumoniae . The immune response to these bacteria is typically Th 1 -dominated .

</thought>

<answer>

B . It primarily induces the Th 1 -cell response . </answer>